REGIONAL CLIMATE & DISASTER POLICY TRACKER

The team handbook

Method, scoring and dashboard priorities

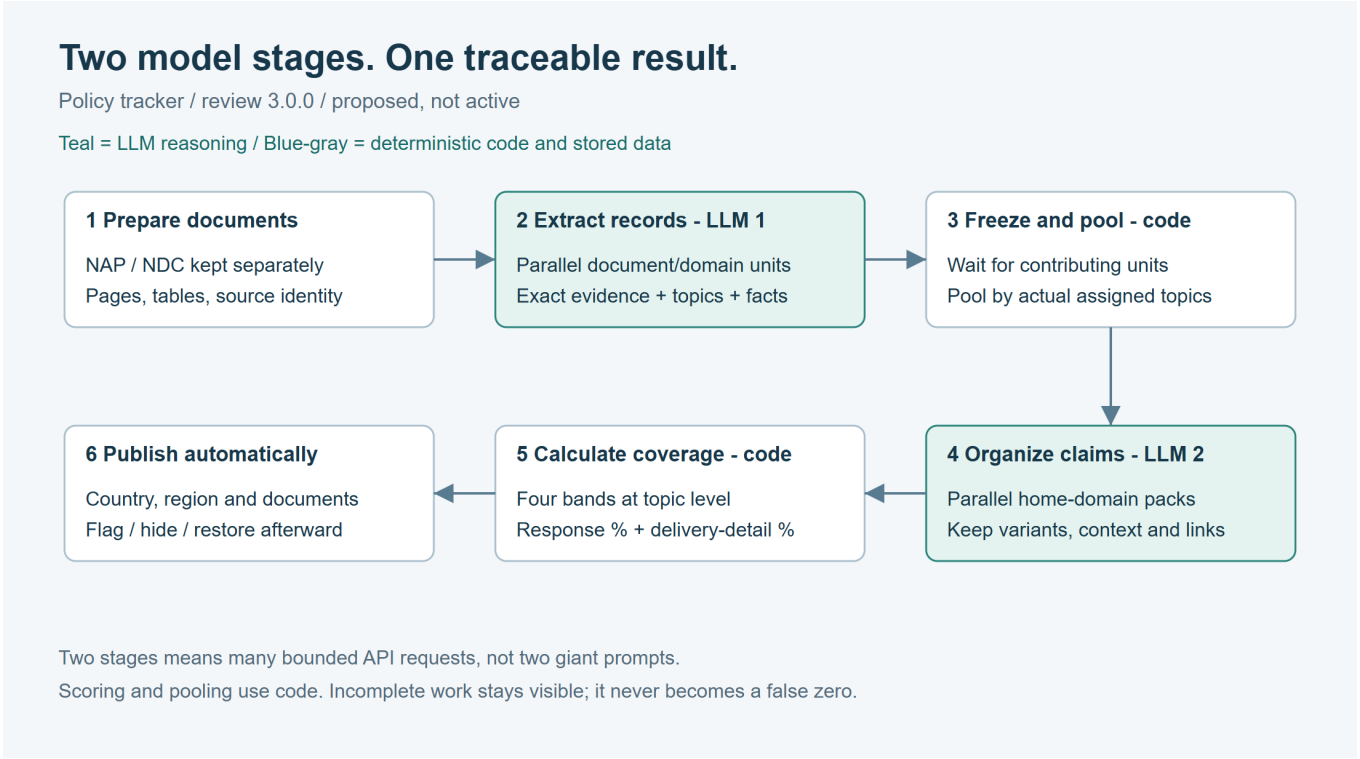
19 October 2026 · Review specification 3.0.0

Use this book for a team discussion. The main flow is two LLM stages with code-owned pooling, evidence checks, scoring and publication. Topic bands describe source documentation, while country/domain percentages count common topics.

The chapters cover the end-to-end process, numerical scoring, screen priorities and failure cases. Technical details can be skipped on first reading. PDF bookmarks navigate chapters and sections.

Reading path: process diagram → scoring example → dashboard priorities → deeper audit. No recurring manual moderation gate is proposed.

The process at a glance



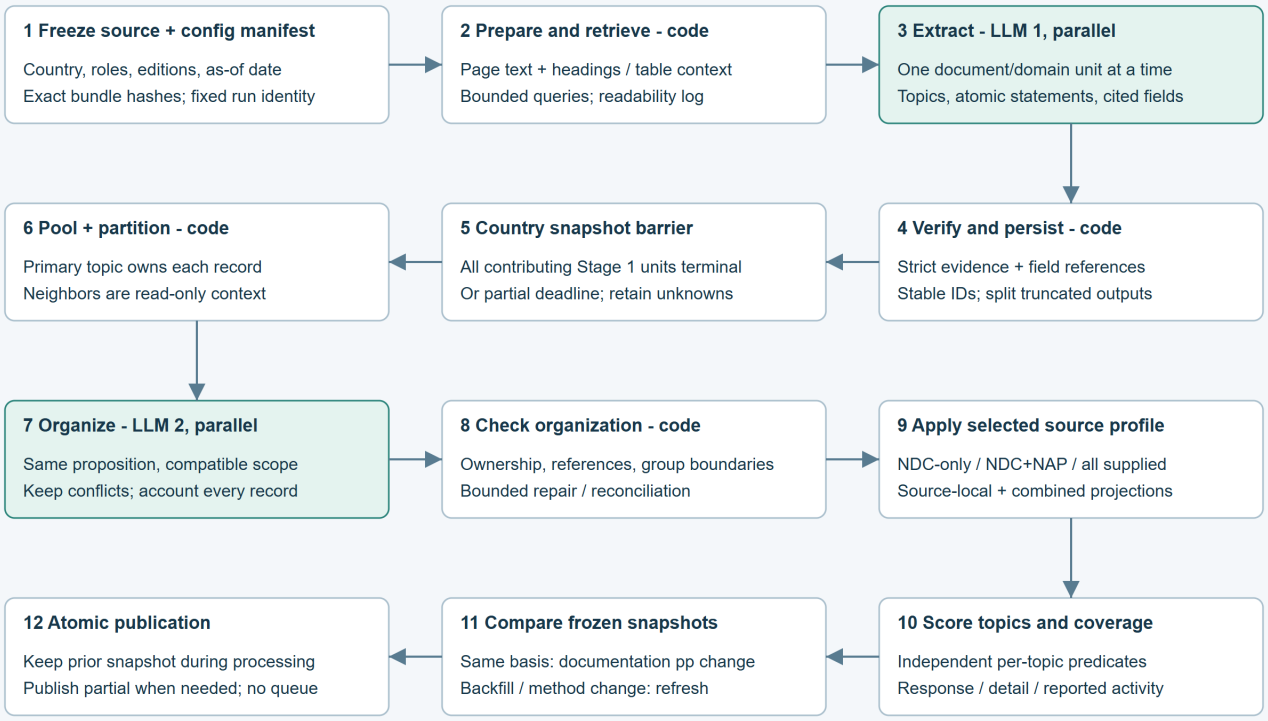
Two logical LLM stages run as bounded tasks. Independent extraction units run in parallel. Organization starts only after the contributing input snapshot is frozen, then independent packs can run in parallel.

Engineering flow

From supplied documents to comparable coverage

Engineering flow / review 3.0.0 / no human acceptance gate

Teal = LLM reasoning / Blue-gray = deterministic code and stored data



MODEL ALLOWLIST: prompt + enabled vocabulary + strict schema + authorized source / record units only.
No team guides, archives, previous scores, unrelated files or arbitrary folder scans enter model context.
One shared cost allowance. Late success creates a newer snapshot; stale jobs cannot overwrite it.

Pipeline: simple operation, traceable evidence

19 October 2026 · Version 3.0.0 · **Review specification, not active application behavior.** The executable reference builds requests and checks data offline. It makes no LLM calls, changes no database and activates no new runtime behavior.

The recommendation in one minute

Keep **two LLM stages**. Stage 1 reads each selected document through the existing domain questions and extracts sourced statements. Code collects those statements by their assigned topics. Stage 2 organizes related assertions across the country's documents. Code then assigns the four documentation bands and publishes the dashboard automatically.

The first operating profile uses **NDC and NAP documents**, within the existing regional climate/disaster scope and **16 domains / 64 topics**. Additional document roles can be introduced through an explicit source-profile extension. Keeping disaster topics does not require pretending that a missing DRR document was supplied.

```
flowchart LR
  A[Selected NDC / NAP files] --> B[Code: pages, retrieval and coverage]
  B --> C[LLM 1: sourced records]
  C --> D[Code: check, retain and pool by topic]
  D --> E[LLM 2: claims, variants and relationships]
  E --> F[Code: bands and coverage]
  F --> G[Automatic dashboard snapshot]
```

Pooling and scoring do not call another LLM. There is no mandatory human review or validation gate before publication. Users can inspect, flag or hide published material afterward. Failed processing remains visible as incomplete, rather than becoming evidence that a policy is absent.

Step by step

Step	What happens	Simple example
1. Select the source set	Code freezes document IDs, roles, editions/date rules and the configuration version. Historical files remain accessible separately.	The current comparison uses a selected NDC and NAP. Excluding an older edition from this view does not mean the policy was withdrawn.
2. Prepare evidence	Extract/index readable pages. Retrieve bounded domain/topic query bundles and retain necessary section headings, table headers and footnotes as separately identified source units.	A target row on page 18 refers to the table header on page 17. Both units are supplied and can be cited.
3. Extract records	LLM 1 returns atomic statements, taxonomy choices and exact source quotations. Facts and delivery details are populated only when supported.	“Water is a priority sector” becomes context with explicit unranked priority wording. It is not automatically a commitment.
4. Check and retain	Code checks schema, known IDs, source scope, exact quotations and field provenance; assigns durable IDs; retains source-local assertions.	“10-20 ha” cannot pass as “1020 ha”; a minus sign cannot disappear.
5. Pool by assigned topics	Code reconciles true duplicate discoveries and gives each record one owner through its primary topic. Relevant records owned elsewhere become read-only context.	A water record found through an agriculture question still reaches its assigned topic pool.
6. Organize common claims	LLM 2 proposes groups, compatible variants and typed relationships using records and quotes. It has no PDF/web access.	Equivalent NDC/NAP targets share a claim; contextual capacity remains separate and linked.
7. Check accounting and calculate	Code checks each owned record appears once, links resolve and source scope is preserved. It computes documentation bands from eligible evidence.	A plan's actor, timeframe and specific deliverable must belong to the same response.
8. Publish automatically	Create a coherent snapshot after jobs terminate or the configured partial deadline. Later success creates a newer snapshot.	A documented response remains visible even if another document is unreadable. The unresolved work receives a separate coverage marker.

A multi-document example

Assume the documents state:

1. NDC: efficient irrigation will cover **100,000 ha by 2030**.

2. NAP: the comparable target is **80,000 ha by 2030**, with the same scope and conditions.

3. NAP: existing irrigation capacity is **60,000 ha**.

4. NAP: the Ministry of Agriculture will rehabilitate **30 km of canals during 2026-2030**.

The output keeps a target claim with two conflicting variants, a separate capacity claim and a separate rehabilitation response. The capacity can provide context, and the rehabilitation plan can support the target, without proving that the target was achieved. Different years or conditional pathways would not automatically

be contradictions.

The combined topic can show **Delivery details documented** because the rehabilitation response identifies an actor, action timeframe and specific deliverable. The target conflict remains visible. A document-only view uses only that document's own evidence; the NDC does not inherit the NAP's delivery details. Historical views exclude later documents.

Precisely what the LLM receives

The actual editable instructions are extract.md and organize.md. retag.md is maintenance guidance, not a routine third stage. The offline builder currently implements the first two stages only.

Input	Stage 1	Stage 2
Prompt	Extraction System instructions and User input sections	Organization System instructions and User input sections
Taxonomy	All 64 topic IDs with concise definitions, plus current/adjacent detail and routing	Relevant domain/topic definitions and relationship vocabulary
Vocabulary	Only enabled facets and required statement/fact/delivery choices	Relationship and comparability choices; record labels remain attached
Source data	Explicit physical-page units from one selected document	Immutable selected records, exact evidence, facts, owned/context-only status and prior identity candidates
Output schema	Strict raw record/evidence schema generated from active taxonomy/labels	Strict claim/variant/link proposal schema; references rather than copied facts
Model access	Supplied source packets only in this reference; retrieval belongs to code	No document search, web tool or directory access

The builder reads exactly five allowlisted files per stage: that stage's prompt plus taxonomy.json, labels.json, pipeline.json and llm-contracts.json. It compiles the relevant pieces; it does not dump all five files into the prompt.

No LLM reads this handbook, the team documents, preserved runtime copies, archives, example fixtures, scoring rules, dashboards, environment credentials or the entire project. The two synthetic examples are passed explicitly by the demonstration command; ordinary request building never searches for them. Titles, filenames, URLs and source text remain data even when they contain apparent instructions.

Source-profile completeness is computed from the selected inventory and required roles. Search/processing completeness belongs to the job/source manifest. Neither is a model output. A missing required NAP remains a source-profile gap even when the supplied NDC was fully processed.

An explicitly exploratory supplied-all profile can set required roles to an empty list. This does not certify matched regional comparison. The current source manifest and cutoff remain external provenance: changing a country snapshot ID does not change an unchanged document's extraction prompt. Stage 2 receives only metadata for documents represented in that pack.

Compact model output versus stored data

Keep model output smaller than the database model. Stage 1 returns local references, statement/type/topic choices, evidence quotes, optional facts, nullable delivery details, enabled facets, explicit priority wording and field provenance. It does **not** manufacture global claim/policy IDs, verification flags, scores or completion states.

`field_evidence` is a fixed-shape array of JSON Pointer paths and local evidence references. Code can turn it into a stored map after validating the references. Unknown values stay null or the vocabulary's not_stated value. Empty optional arrays are allowed; additional labels do not earn a higher band.

Stage 2 returns local proposed claims/variants, supplied record IDs, proposed detail associations and record-to-record relationship edges. It references immutable facts instead of copying them again. Context-only evidence can be linked without becoming a second owned record. Unmapped records remain visible in a local code-retained bucket and can be relabeled after taxonomy changes.

Code assigns persistent identities, source-page mappings, versions, budgets, automatic check states and documentation bands. It can check referential consistency; it cannot prove every semantic association merely because IDs and JSON are valid.

If organization fails or retains an unresolved record, code can publish its verified source-local assertion as a **provisional singleton assessment**. Directly evidenced responses and delivery details stay visible, with no cross-record borrowing. Mark organization incomplete; missing predicates remain unknown and common-claim counts remain provisional rather than a headline total. Later successful organization publishes a newer snapshot with identity lineage. This fallback needs no extra model or human review.

See DATA-DICTIONARY.md for raw-to-stored field mappings and the complete field options. The exact compiled schemas and resolved prompt strings are visible in extraction-request.json and organization-request.json. Both are explicitly synthetic previews, not actual country runs.

Parallel work, ordering and limits

Use an initial configurable maximum of **four concurrent country work units**. Document/domain extraction units run in parallel. Freeze the selected source/configuration manifest and wait until **all** Stage 1 units in that manifest are terminal-or the partial deadline is reached-before pooling. Then organize independent home-domain packs in parallel.

This barrier must include all contributing extraction domains. Waiting only for the same domain question is insufficient because another question can discover a record assigned to this topic. Coalesce uploads into one update rather than repeatedly regenerating the same group after every arrival. Later unrelated uploads belong to a newer manifest.

Initial setting	Target	Hard guard / behavior
Stage 1 supplied source text	8,000 estimated tokens	18,000 estimated source tokens; split larger packets
Stage 1 output reservation	6,000 tokens	On truncation, split source units; do not resend an unchanged oversized task
Stage 2 record/context input	12,000 estimated tokens	24,000 estimated record/context tokens; partition and reconcile boundaries
Stage 2 output reservation	6,000 tokens	On truncation, split owned-record batches and preserve membership accounting

Initial setting	Target	Hard guard / behavior
Retrieval per document/domain unit	Up to 3 search calls, up to 20 results each	Log queries, returned source units, omissions and unreadable material
Interactive publication deadline	60 minutes	Publish explicit partial state; this is not a guaranteed completion time

These are **unmeasured initial operating hypotheses**, not promised efficiency or accuracy. The offline builder uses characters divided by three as a conservative working estimate and rejects requests above the configured guard. It never silently drops input. This estimate is not a tokenizer: production must check the selected model's actual tokenization/context rules, including instructions, schemas, source/context, output/reasoning and safety margin.

Source/table context can be carried into adjacent partitions without assigning it duplicate record ownership. Boundary identity checks remain within Stage 2. A chain of similarities must not merge incompatible endpoints: A resembles B and B resembles C does not establish that A and C describe the same target.

Where efficiency comes from

Two stages do not require two different models. Start with the same validated model for both; test a cheaper organization model as an optional override only if it preserves grouping, contradiction and source-attribution quality. Organization can be harder than extraction, so a cheaper default is not automatically safer or more efficient after repairs.

• Reuse extraction for unchanged document bytes and semantic configuration. Reorganize only affected topic groups and dependency neighbors.

• Keep all topic IDs available, but avoid repeatedly sending unrelated full operator documents or disabled vocabulary modules.

• Skip an LLM call when a completed code retrieval plan returns no source units; record the bounded-search outcome and limitations in code. Skip empty organization packs. Do not infer whole-document absence from an empty extraction response.

• Return record/evidence references from Stage 2 instead of repeating large factual objects.

• Recalculate a band-rule edit from retained facts. A display-name edit needs no model call. A genuine topic expansion needs new source search, not only retagging.

• Use one persisted country-update cost allowance across search, retries, repairs, OCR and overflow. Keep uncertain spend reserved; report storage separately. Do not multiply independent SDK and worker retry budgets.

Extraction cache keys use document/source-unit content and semantic prompt/schema/model settings. Organization keys use owned/context record content, relevant document metadata and identity candidates. Country snapshot IDs do not invalidate either by themselves. A source cutoff that actually changes selected evidence changes the dependent pack, as it should.

Retain the 16-domain retrieval approach initially. Compare it against code-deduplicated retrieval packets and, later, section-first extraction on the same representative sources. A section-first approach may reduce repeated context but may also lose targeted topic coverage and produce larger outputs. No default change is justified by call count alone.

Concrete source-code findings to resolve during integration

These references describe the current application, not fixes already activated by this review package.

Existing behavior	Planned correction
verify.ts:23 removes dashes/minus signs; verify.ts:43 also removes whitespace in a matching path. “10-20” can therefore match “1020”, and a negative figure can match a positive figure.	Strict eligibility checks preserve signs, ranges and comparators. Tolerant highlighting is a separate operation. The offline reference tests reject both examples.
synthesise.ts:15 strips source quotes/topic tags; synthesise.ts:49 selects the same extraction-domain answers.	Pool by assigned topics; retain evidence and qualifiers in the organization input.
handlers.ts:46 waits for extraction jobs in the same domain.	Freeze the full selected input manifest and account for all contributing Stage 1 units before pooling.
call.ts:28 retries a failed/truncated task; openai.ts:243 defaults to a large output allowance.	Explicit output reservation and deterministic splitting; never repeatedly retry the same oversized payload.
openai.ts:251 limits search results, not total search calls; cost.ts:41 supplies an estimated cap.	Code-owned query accounting and one persisted runtime update allowance.
schemas.ts:20 has only the older limited figure contract; source-linked delivery checks are absent.	Implement the reviewed raw schemas and explicit code-owned normalization adapter before activating the four-band evaluator.

Run the reference locally

From the repository root, using Node 24 or later:

```
node docs/review/reference/build-inputs.mjs --write-examples
node docs/review/reference/input-contract.test.mjs
```

The first command regenerates only the two synthetic request previews. Each includes resolved prompt text, a compiled strict schema, exact input-file SHA-256 hashes, selected source IDs and clearly labeled character-based estimates. A production dispatcher would add the selected supported model; the offline builder does not read credentials or call a provider.

The tests exercise allowlist rejection, unknown IDs, prompt-placeholder handling, untrusted source instructions, source-scope isolation, missing required roles, full-catalogue routing, context/ownership separation, exact record accounting, field provenance, cropped numeric signs/ranges/comparators, as-stated units/periods, stable incremental requests and overlarge input failure. The numeric-span guard recognizes common numeric and English comparator forms; it is not a universal multilingual quantity parser. These tests do not measure real-document extraction accuracy or prove semantic entailment or prompt-injection resistance of a model.

Before runtime activation, integrate the schemas, storage adapter, scheduler barrier and budget ledger; evaluate representative readable, scanned, tabular and multilingual NDC/NAP documents. Keep the residual limitations explicit: retrieval can miss evidence, table/OCR interpretation can fail, and a correct quotation can

still be misclassified. These development checks do not create a country-by-country human approval queue.

Scoring review v3: coverage numbers with clear topic detail

Version 3.0.0 · review_not_active. This is an editable recommendation and executable reference specification. It is not the running application's scoring system. Rules are in scoring.json.

The recommendation in one minute

Use **two main percentages**, supported by topic colors and evidence. Keep implementation reporting separate.

Measure	Question	Example in a ten-topic domain
Response coverage	How many common topics document a policy response?	6/10 = **60%**
Delivery-detail coverage	How many document Who + When + a specific What for at least one coherent response?	2/10 = **20%**
Implementation reported, secondary	How many explicitly report an identified response underway or delivered?	1/10 = **10%**

Do not blend these percentages. The first describes breadth, the second specification, and the third reporting. None measures adequate ambition, actual effectiveness, resilience or national effort. A detailed plan is not implementation; an implementation report need not contain all three delivery details.

At country level count **unique common topics**, not domain averages: “32/64 topics document a response - 50% coverage.” This is a documentation inventory statistic, not a national policy-quality score. There is no headline claim score.

1. Keep four topic colors

Code	Exact label	Meaning
0	No evidence found	Complete processing of the defined usable-source scope found no qualifying topic evidence
1	Context documented	Concern, context, projection, reported effect or unadopted proposal, without an attributable policy response
2	Policy response documented	At least one attributable commitment, planned action, implementation assertion or delivered output
3	Delivery details documented	At least one coherent response has accountable actor, action timeframe and a specific verifiable deliverable
null	Unavailable	No qualifying evidence is available and processing is incomplete

The code is an ordered category, not a quantity. Do not map the codes to 0/33/67/100 or average them. Topic colors use this legend. Domain/country percentages use neutral bars and exact counts, without inventing additional good/bad thresholds.

A limitation to make visible: one well-specified project can establish topic band 3. It does not prove complete policy coverage. Show response count, detail-ready response count and conflicts beside the topic. A detail drawer can show “2 of 3 delivery details documented” for a selected response, but this is not a new topic rating.

2. Calculate delivery details from three checks

Check	Qualifies	Does not qualify
Who?	A named responsible or implementing organization	Funder, consulted body, supporting organization or merely mentioned actor alone
When?	An action period, delivery deadline or dated milestone	Publication date or unrelated observation year alone
What specifically?	A quantified output, identified asset, named legal instrument or identifiable milestone	“Improve resilience” or generic “rehabilitate canals”

All three must refer to **the same coherent response**. “The Agriculture Ministry will rehabilitate 30 km of canals by 2030” qualifies. “The ministry will rehabilitate canals by 2030” lacks a sufficiently specific deliverable.

A qualitative deliverable can qualify: “Adopt the National Drought Management Regulation” does not need an invented quantity. Conditional commitments count as responses; show the condition separately. Advice and unadopted recommendations remain proposals, even when detailed.

Compatible documents can supply different details of the same identified response. A separate supporting project cannot donate its actor, deadline or output. The model extracts the facts; code applies the rubric. No LLM scoring call is needed.

Secondary-topic credit also needs direct evidence for each predicate. A topic tag alone cannot inherit the primary topic's response, delivery or implementation status. The reference adapter supplies explicit per-topic projections when a claim serves multiple topics.

3. Worked irrigation example

All data in water.json are invented for demonstration. The combined example is a `supplied_all` exploratory inventory containing both NAP editions, not the matched current NDC/NAP corpus.

Source assertion	Interpretation
NDC: 100,000 ha of efficient irrigation by 2030	Policy response, but no accountable actor stated
NAP: comparable target of 80,000 ha by 2030	Conflicting target variant; retain both values
NAP: existing network served 60,000 ha in 2024	Separate context; not 60% achievement

Source assertion	Interpretation
NAP: ministry will rehabilitate 30 km of canals during 2026-2030	Separate planned response with delivery details

The combined topic is **Delivery details documented**, because the canal action qualifies. Its target claim remains unresolved and cannot borrow the canal project's actor. No implementation is reported.

In this deliberately tiny **one-topic example profile**, response coverage is 1/1 = 100%, delivery-detail coverage is 1/1 = 100%, and implementation reporting is 0/1 = 0%. This means that the one checked topic contains those kinds of documentation; it does not mean perfect irrigation policy. The real regional profile has its full fixed topic denominator.

The NDC-only view stays band 2; the 2024 NAP is band 1; the 2026 NAP is band 3. Document evidence and stages never leak backward into earlier editions. The normalized input, calculated output and reproducible checker make the calculation inspectable.

4. Treat ambiguity separately for each measure

Each coherent claim variant yields three yes/no facts: response present, delivery details present, implementation reported. If variants disagree about a fact, preserve possible outcomes.

• Claim lower: every coherent alternative qualifies.

• Claim upper: at least one alternative qualifies.

• Topic lower: at least one claim definitely qualifies.

• Topic upper: at least one claim might qualify, or required processing is incomplete.

For each measure separately, count:

• **P:** topics where the fact is demonstrated.

• **A:** topics where complete eligible processing establishes it was not found.

• **U:** topics where it remains undetermined.

• **N = P + A + U:** the fixed common denominator.

Coverage is **100 × P/N** when U = 0. Otherwise show **100 × P/N to 100 × (P+U)/N**.

Example: six responding topics, two complete topics without a response, and two unknown topics give **60-80%**, not 75% after silently dropping unknowns. Processing completeness appears separately: if those eight determinate topics were all processed, it is 8/10. P+A is not automatically the processing count.

An established response remains positive if another source fails. Delivery details and implementation can still be unknown. Thus response coverage can be exact while the corpus is incomplete; this does not make the snapshot eligible for a complete comparison.

A conflict over 80,000 versus 100,000 hectares does not automatically change any predicate. A conflict over whether a measure is only proposed or actually undertaken creates response uncertainty. Planned versus underway alternatives create implementation-report uncertainty while response coverage may remain exact. Never resolve disagreement by taking the most favorable alternative.

5. Preserve implementation and history

Use source statement types for reported stages: Commitment, Planned, Underway, Delivered. Context and projections have no implementation stage. A reported outcome alone does not prove a policy caused it.

The secondary implementation metric counts explicit `implementation_underway` or `output_delivered` assertions. It does not require band 3, and band 3 does not establish it. This is **implementation reporting in the selected corpus**, not a measurement of actual achievement.

Alternative variants in a current assessment are coherent competing interpretations, not an unfiltered timeline. Historical assertions remain in history; explicit update lineage determines which assertions belong in a current comparison. Do not use a historical maximum stage as current status. A newer source edition alone does not prove that an older policy has been withdrawn.

6. Compare the same documentary scope

The source profiles are editable, versioned definitions inside `scoring.comparison.source_profiles`:

Profile	Source selection	Use
`supplied_all`	All explicitly selected supplied documents	Exploratory inventory; no comparable trend arrow
`ndc_only`	Latest eligible NDC edition in the agreed family as of the cutoff	Matched NDC-documentation comparison
`ndc_nap`	Latest eligible edition for each required NDC and NAP role/family as of the cutoff	Matched NDC/NAP-documentation comparison

A profile chooses the comparison corpus; **it does not establish policy supersession or withdrawal**. Historical editions remain accessible outside the selected current corpus. Carry-forward of an older assertion needs explicit supporting evidence and provenance. It is not silently counted forever.

Missing a required document role means `source_profile_complete=false`. The reference calculator requires `scope.source_profile_complete` explicitly. Input topic `processing_complete` records whether the supplied jobs finished; the output's effective `processing_complete` requires **both** those jobs and the required source profile to be complete. `job_processing_complete` and `job_processed_topics` preserve technical-job success separately. With a missing required NAP, an empty successful NDC search remains unavailable/unknown, not zero. Positive evidence is retained. Never automatically substitute NDC material for a missing NAP. A functional equivalent needs an explicit versioned rule applied consistently.

Store exact source IDs, roles, family IDs, edition dates, cutoff date and ingestion timestamps. A source's publication/edition date is different from when it was uploaded. Late uploads can improve the inventory without representing a new policy.

The existing 64-topic profile is a common inventory, **not 64 universal national policy obligations**. DRR topics absent from an NDC/NAP corpus mean not found there; they do not establish weak national disaster policy. A DRR-specific comparison needs a shared profile including appropriate disaster-policy sources. Implementation comparison may also need progress-report sources because planning documents are not comprehensive implementation reports.

7. Country priorities and fair regional interpretation

Country A may explicitly prioritize drought and water management; Country B may emphasize resilient tourism. Highlight these stated priorities with citations. They can sort or filter the dashboard, but do not silently change common topic weights or denominators.

The same water coverage percentage means the same amount of **topic presence in the matched documents**. It does not mean equal ambition, urgency or effort in those different contexts. Do not penalize a country for failing to discuss a subject simply because another country calls it a priority.

Do not infer N/A from silence or simple geography. Landlocked countries can have fisheries; inland countries can have flood-protection policy; countries without glaciers can depend on transboundary mountain water. A future narrower common regional core is a configurable shared profile, not a per-country self-weighted score. UNFCCC's NAP guidance emphasizes country-driven, non-prescriptive planning; our priority lens preserves that context alongside shared documentary comparisons.

At national level count each common topic once. If a small domain has 2/2 response topics and a large domain has 0/8, national inventory coverage is **2/10 = 20%**, not the unweighted domain average of 50%. No country quality composite or z-score is introduced.

Regional averages and standard deviations

The most useful regional topic statistic is **prevalence**, not an average band: “7 of 10 comparable countries document a water-supply response.” Calculate response and delivery-detail prevalence separately, with an unresolved range when needed. Show how many countries satisfy the source profile and how many are excluded or incomplete.

For a domain, an optional distribution of country coverage percentages can show country dots and the cohort median. Use only the same topic basket, source-role/date rules and method. State the cohort and denominator; do not silently exclude countries to improve an average. If the cohort is small, the individual dots are more informative than a distribution summary.

A mean and standard deviation of **coverage percentages** can be an advanced descriptive export for complete comparable cohorts. They describe dispersion of documentation coverage, not effort. Do not calculate them from ordinal band codes, assign arbitrary midpoint scores to unknown intervals, or turn them into a z-score country ranking. Standardizing against a weakly documented region can make a modest document look excellent while concealing the absolute gaps. Keep these statistics out of the initial headline and retain absolute percentages as the shared reference.

8. Use arrows only for comparable documentation changes

A 60% → 70% change is **+10 percentage points of documentation coverage**. It is not a 10-point improvement in climate resilience or proof of implementation progress.

An arrow requires the same country, frozen extraction/method signature, taxonomy and scoring versions, unique topic homes/profile, source-role/family coverage and date rule. Both snapshots must be recomputed on that same basis, have complete required sources and topic processing, and have an exact value for the chosen measure.

Event	Default display
Newer eligible edition of the same source family, after the old cutoff, on the same complete basis	Neutral up/down/unchanged documentation arrow and percentage-point difference

Event	Default display
Ordinary upload, backfill of an older source, source removal or recalculation	Refresh annotation; no progress arrow
Method or topic/source profile change	New/recomputed baseline; no direct arrow
Partial processing or unresolved qualification	Arrow unavailable; no point delta
Evidence omitted in a later edition	“Not restated in current corpus”; no inferred withdrawal
Explicit withdrawal or supersession	Preserve source-backed annotation and lineage

For specialists, an optional possible difference is **[new lower − old upper, new upper − old lower]**. This is an uncertainty range, not a statistical confidence interval. Default UI still suppresses point arrows for partial snapshots.

Use neutral direction icons for both increases and decreases. Keep four-band topic colors separate from these changes. See DASHBOARDS.md for matching screens.

Official SDG trend assessments concern indicator trajectories toward targets. The visual clarity is useful here, but this documentary tracker uses a different construct and must not call coverage “on track” or “target achieved.” UNSD technical note.

9. Options and simple SWOT

Option	Strength	Weakness	Opportunity	Threat
A. Separate coverage percentages + four topic bands - recommended	Literal counts; easy to reproduce	No smooth topic-quality number	Clear regional gaps and evidence drill-down	Users may overread coverage as adequacy unless labels stay explicit
B. Topic documentation-checklist percentage	More granular-looking topic score	Weights are design choices; one project can satisfy the whole checklist	Useful later if fixed topic-specific checks are agreed	False precision and maintenance burden
C. Percentage of extracted responses with full details	Describes the observed response portfolio	Denominator depends on extraction, splitting and corpus size	Useful diagnostic within a country	Richer documentation can lower the score; unreliable league tables

The recommended design removes hidden weighting and keeps disagreements, priorities and source gaps visible. It can be implemented without a third LLM stage. No method automatically converts policy prose into verified policy effectiveness.

10. Reference implementation and its limits

score-bands.mjs consumes normalized, already source-checked assessments. It reads the editable scoring configuration. band-cases.json contains full synthetic inputs, while score-bands.test.mjs tests ambiguity, missing evidence, identity, duplicates and the separate metrics.

The core input is `scope`, a fixed topic `profile`, resolved `evidence`, coherent `claims[].variants`, and `topics[].processing_complete`. A variant includes `policy_object_id`, statement type, conditionality, its admitted evidence IDs and the three delivery details. All details are explicitly present or null. Detail citations must belong to that coherent variant. Each variant has one claim owner; all variants of a claim share the same policy-object ID.

For more than one topic, `claims[].topic_projections` is required: one `{topic_id, variants:[...]}` projection per topic. Each retains all canonical variant IDs, but includes only statement type and delivery qualifications supported for that topic. It may downgrade a response to contextual evidence and omit delivery details; it cannot invent new evidence, a stronger response stage or new delivery qualifications. The engine computes topic metrics from these projections rather than inheriting the primary claim band. Source relevance is still checked upstream. Single-topic inputs need no projection wrapper.

Outputs expose `topics[].metrics`, `domains[].metrics` and, for country scope, `national.metrics`. Each metric contains its own P/A/U/N, lower/upper percentage and exact flag. Existing response-coverage fields remain aliases for compatibility. Internal claim band codes support diagnostics only; they are not headline UI ratings.

The seven allowed actor roles are responsible, implementing, financing, supporting, monitoring_reporting, consulted and unspecified; only the first two satisfy Who. Timeframe anchors are action or publication; only action qualifies. Deliverable kinds are quantified_output, named_asset, named_legal_instrument, named_milestone and generic_action; generic_action does not qualify.

`assessDataset` rejects malformed inputs. `assessDatasetSafely` returns unavailable with a null assessment on invalid input; it never silently substitutes zero. “Available” means calculable, not complete. The engine cannot prove source truth, semantic support, exhaustive search, correct policy identity, authoritative topic homes or raw-record ownership. Those remain upstream automatic-check and evaluation responsibilities.

compare-snapshots.mjs separately checks frozen basis metadata, exact selected source manifests, cutoffs, newer same-family editions and completeness before permitting a documentation arrow. Its test cases cover refresh events and suppression. Metadata checks are not proof that a source was correctly identified.

Run from the repository root:

```
node docs/review/reference/score-bands.test.mjs
node docs/review/reference/compare-snapshots.test.mjs
node docs/review/reference/check-water.mjs
```

Changing display labels is a presentation edit. Changing qualification rules, topic scope or source profiles requires a versioned recomputation. Reuse stored facts when they establish the new rule; otherwise re-extract only affected material. There is no human approval queue before publication: bounded automated checks finish with either usable results or explicit incomplete states. Users may flag/hide results afterward; analytical exclusions create clearly labelled adjusted snapshots.

Dashboard review: what users see and do

Review specification 3.0.0 - not active in the application. The dashboard describes the supplied climate and disaster policy documentation. It does not rate actual resilience, policy effectiveness or the adequacy of ambition.

The main path is **Region** → **Country** → **Domain** → **Topic** → **Claim** → **Source**. Keep that path visible in breadcrumbs. A document selector changes the evidence scope without creating another navigation system.

One visual language

Visual element	Meaning
Response coverage %	Share of the fixed topics with an attributable policy response documented.
Delivery-detail coverage %	Share of the same fixed topics with at least one response specifying a responsible/implementing organization, action timeframe and verifiable deliverable.
Implementation reported - secondary	Topics with source-reported implementation or delivered output under the evidence rules. This is not verified success or target achievement.
Four topic-band colors	No evidence found; Context documented; Policy response documented; Delivery details documented.
Hatch and status text	Incomplete processing or unresolved qualification. Unknown does not become zero.
Conflict / condition / priority badges	Different source properties, shown separately from the bands.
Change arrow	A documentation change between comparable complete snapshots, with evidence behind it.

Use at most **three headline numbers**, led by the first two. Prefer percentages with a visible fixed denominator and an explanation on selection. Band distributions explain those numbers. Raw finding counts, quote-match rates and model costs belong in evidence or run details.

Topic bands use one ordered neutral-to-blue palette with readable labels. Domain icons may use distinctive accent colors. **Do not color an entire domain or country by its maximum topic band, average band code or mean implementation stage.** Unavailable has a neutral patterned treatment outside the four-band scale.

Country percentages count each of the **64 unique topics once**. They are not means of the 16 domain percentages: domains have different numbers of home topics. Domain percentages use their fixed home-topic subsets. Four bands are categories, not numerical weights.

For each coverage measure, count **P** topics already established, **A** topics where qualifying evidence is absent after resolution, and **U** unresolved topics; **N = P + A + U**. Show `100 × P/N` to `100 × (P+U)/N`; use a single percentage when U is zero. For example, six established, two absent and two unresolved topics give **60-80%**. Apply the rule separately to response, delivery details and implementation reported; their unresolved sets may differ. Processing completeness remains a separate status, not `N−U`. Show supplied-job completion and missing required document roles separately. Effective completeness requires both: a missing required NAP keeps unproved coverage unknown even when every supplied file finished.

Implementation reported requires an assertion of the identified response being underway or an output delivered. It does not require the topic's delivery-detail band, and the delivery-detail band does not establish implementation. A planned project can specify delivery very clearly.

1. Regional overview - first priority

First look: show the region, selected countries, as-of date, comparison profile and source eligibility before the matrix. The default profile must name its document roles: for example, **NDC only** or **NDC + NAP** with the same edition/date selection rule for every country. All-supplied-document exploration remains clearly labelled exploratory.

The country × domain matrix displays response coverage as the primary number. A small distribution below it shows how the topics fall across the bands; delivery-detail coverage can appear on hover/selection or in a metric switch. Incomplete cells display ranges rather than manufactured single values. Do not try to fit every badge and metric into every cell.

Drill-down: selecting a domain opens a country × topic matrix. Topic cells use the four bands directly. Selecting one opens the same topic drawer used elsewhere, with the country and source scope retained.

Actions: select countries, domain, source profile or as-of date; compare up to a few countries side by side; export a table with band names, coverage and source references. Default order is alphabetical or the user's chosen order, not a national quality ranking.

Example: “Water: 4 of 6 home topics document responses” can appear beside “2 of 6 specify delivery.” The comparison is meaningful only within the same Water topic basket and source profile. A country with an NDC and NAP is not silently compared with another country represented by its NDC alone.

2. Country overview - first priority

First look: a short source header shows included NDC/NAP editions, as-of date, processing status and comparison eligibility. Then show the two headline coverage numbers; implementation reported is a smaller secondary indicator. Keep all 16 domain tiles available.

Each domain tile contains its name, home-topic denominator, response coverage and a small band distribution. Delivery-detail coverage may be the second number. Move detailed counts, stage distributions and technical diagnostics into the selected domain. One well-documented project must not make the whole tile look complete.

Drill-down: selecting a domain lists its home topics with band, one-sentence explanation and any conflict/incomplete badge. Related topics from another domain appear in a separate section: **“Also relevant - counted under another domain.”** AGR-01 irrigation can be visible under Water while remaining counted under Agriculture.

Actions: open topics, select documents, compare snapshots, add documents and export evidence. Use stable navigation-**Overview, Topics, Documents**-rather than adding a tab for every uploaded document. Put any generated brief inside Overview; it must not become another competing result system.

Remove the default radial chart of finding counts. Finding volume depends on extraction and splitting choices; a larger wedge can look like stronger policy when it merely means more text.

3. Topic drawer and claim detail - first priority

First look: topic name and code, band, a short reason, and the source scope. For band 3, use “At least one response has delivery details,” not “Topic fully covered.” Composite topics may contain several policy aspects.

Show these sections in order:

1. **Documented responses.** Common claims, their statement types and the three delivery checks: who, when, what specifically. Each check opens the exact source evidence for that response, variant and selected topic. A secondary topic only receives its own source-qualified response/details; it never inherits all checks

from a tag.

2. **Concerns and context.** Risks, needs, baselines and reported effects. Keep them visible without implying implementation.

3. **Disagreements and related evidence.** Target variants together; capacity and supporting actions as separate linked claims. Explain the relationship in one sentence.

4. **Sources and limitations.** Document/page links, searched scope, incomplete work and configuration version.

Actions: open a quotation on its physical page, compare documents, inspect history, flag a claim, hide/reveal it and export. Technical field provenance belongs under “Method details.” Ordinary users should not need to understand prompt variables or schema names.

Example: an NDC's 100,000-hectare target and a comparable NAP's 80,000-hectare target remain conflicting variants. A NAP's 60,000 hectares of existing capacity remains context. A separate ministry plan for 30 km of canal rehabilitation can establish the topic's delivery-detail band without filling the target's missing actor or proving the target has been achieved.

4. Priority and evidence-gap lenses

Provide three plain filters on the topic/evidence view:

Lens	What qualifies	What the user can conclude
Explicit document priorities	Source text explicitly identifies a priority; display that wording and its quotation.	The source prioritizes this concern, need or response.
Needs with no response found	A sourced concern/need and no qualifying response in the current assessed scope. Incomplete work is labelled.	A documentation gap worth investigating, not proof that no action exists.
Conflicting statements	Source-qualified comparable assertions disagree and lack an evidenced resolution.	The documents need closer reading; the tool does not choose a winner.

Priorities are **not** inferred from repetition, severe hazards, document length or model judgement. The optional `priority_evidence` field is source wording, not an importance score. “Drought is a priority risk” remains a concern; it does not become a response.

“Most actionable” should mean these transparent evidence conditions, not an opaque ranking. Users may filter or pin relevant domains, but priority lenses neither drop topics from the benchmark nor change weights. A missing response to a stated need and an analyst's assumed need are different things; show only the former as a documented gap.

5. Documents and source comparison

First look: a document table with title, role, edition/family, included scope, readable/unreadable material and processing state. Select a document from a picker. The selected view uses only that document's evidence and the same topic taxonomy.

Drill-down: a claim-by-document table aligns assertions while retaining each target value, baseline, period, condition and source-local implementation stage. Open the source page beside the table. Use “Not repeated” for an omission; a later document does not automatically withdraw or supersede an earlier undertaking.

Actions: choose two documents, inspect differences, download the source, add a new edition and export the comparison. Retry failed processing from diagnostics. Avoid a prominent “Re-run everything” as the ordinary refresh action; incremental processing should reuse unchanged work.

NDC and NAP are the initial document roles. Additional roles require an explicit profile update and the same rules across the comparison group. Their later availability must not silently redefine a previous benchmark. All-supplied views remain useful exploration, even when country source profiles differ.

6. History and arrows

Arrows describe **documentation change**, not climate progress. A comparable change requires the same topic and source-role profile, method/rubric, date convention, complete source profiles, complete relevant processing and exact metric values. Compare genuine editions in the same document families. Record the before/after snapshot IDs.

• A directional arrow appears only when the comparison establishes a direction under the metric's change rule. It opens the exact changed topics and sources.

• A source backfill or methodology change uses a **refresh symbol** and explanation, not an improvement/regression arrow.

• Incomplete or incomparable snapshots show “Change not comparable” or “Pending,” not a flat arrow.

• Individual claim histories can show a source-supported change from planned to underway. That does not create a domain-wide implementation trend.

The newest NAP versus all earlier documents is not a valid generic progress comparison. A newly uploaded document may reveal older evidence; its upload date must not become the date of policy change.

7. Settings - one editor for the actual bundle

Area	Default controls	Advanced detail
Taxonomy & labels	Domain/topic definition, home, display placements, routing; enabled label modules	Stable IDs, lineage, all enum choices, source-explicit priority field
Prompts	Extract and organize templates	Maintenance prompt, resolved message/schema preview, hashes
Documentation bands	Readable qualification rules and coverage definitions	Versioned changes and regression examples
Processing	Document roles, model roles, budgets and automatic retry limits	Retrieval settings, cache invalidation and publication deadlines

Core labels are purpose, hazards, geographic scope and policy instruments. Themes and affected groups are an optional context module, disabled initially; specialist risk labels are also disabled. The quotes and source assertions remain available. Enabling a module later requires the appropriate automatic retagging or source extraction before its structured coverage can be claimed.

Every edit shows its impact before the user saves a new bundle: display-only; relabel; reorganize; re-extract; or recompute. Keep topic IDs stable when changing display order or an organizational home; semantic splits/merges require explicit lineage. Never use placement order as an implicit scoring-home editor.

This is configuration control, not a moderation screen. Normal outputs publish automatically after bounded machine checks. A saved method change does not turn current results into reviewed or approved policy facts.

8. Unmapped evidence and diagnostics

Provide an **Unmapped evidence** entry with a visible count and searchable source-linked records. Null topic assignment means the subject did not fit; it is not an “Other” domain. A record's statement type stays intact. `statement_type: other` is a separate fallback for assertion type and can occur with a known topic.

Retain relevant cross-cutting climate/disaster concepts even if they lack a topic. Do not auto-add topics from them. Exclude unrelated general development material from the climate/disaster scope. Users can inspect unmapped evidence after publication and later change the taxonomy; there is no review queue holding the dashboard back.

An Activity panel contains costs, actual retrieval/page counts, source-check diagnostics, attempts and failures. Label quote checks as **source matching**, not factual validation. “Readable pages,” “pages retrieved” and “pages cited” are separate quantities.

If organization fails or remains partial, show verified source records as **Provisional individual claims**. Their own documented response/details remain usable; the system never assembles missing details across these provisional records. Mark organization incomplete, preserve uncertainty for missing coverage, and label any common-claim count provisional rather than displaying it as a headline number. Sources remain accessible without a review queue. A later completed run replaces the provisional view in a new snapshot with traceable lineage.

Minimum implementation

Build the regional matrix, country/domain tiles, topic drawer, document comparison, settings and visible unmapped evidence first. Reuse the same evidence drawer across every screen. Support keyboard navigation, meaningful text beside colors, clear incomplete patterns and exports containing evidence IDs.

Post-publication flags/hiding remain recoverable overlays. Hiding affects presentation by default. An explicit analytical exclusion creates a separately labelled adjusted view, leaving the standard regional comparison reproducible. There is no mandatory human approval before publication.

Final end-to-end review

19 October 2026. This audit distinguishes current code, proposed improvements and unresolved empirical questions. Application behavior has not been changed by this review pack. Paths below refer to the repository root.

Findings that justify a change

Priority	Current evidence	Risk	Smallest useful revision
P0	`src/lib/analysis/verify.ts` normalise removes dashes including minus signs; squash removes spaces	A quote can match after changing numerical meaning. Reproduced: -5 versus 5, and 10-20 versus 1020	Strict evidence eligibility preserves signs, ranges, comparators and units. Keep tolerant highlighting separate
P0	`src/lib/analysis/run.ts` stores citation verification flags but can finish a unit with unverified findings; citation arrays can be empty	Schema-valid output becomes apparently complete even without supported evidence	Require supporting evidence for qualifying fields; unsupported material cannot establish a band. Track unresolved processing separately
P0	`src/lib/openai.ts` allows file search; result-count limit does not enforce search execution or complete topic coverage	Empty output can be mistaken for complete absence	Code executes/logs a bounded query plan and owns coverage. A model does not certify that it searched enough
P0	`src/worker/handlers.ts` waits for queued/running work in the same domain	Cross-domain-discovered records may miss the first synthesis	Freeze a contributing country/document/configuration manifest; wait for all its Stage 1 units or the partial deadline
P0	`src/lib/analysis/synthesise.ts` reduces evidence fields and uses document plus array-position references	Organization loses precise context and stable identity; difficult to audit duplicate/contradictory assertions	Keep stable code-owned record/evidence IDs, exact quotes, structured facts and ownership manifests
P0	`src/lib/analysis/prompt.ts` synthesis selects furthest stage; `src/lib/topic-merge.ts` and `src/lib/topics.ts` propagate maxima	A newer action can rewrite an older document's apparent stage or lend detail to a different target	Source-local records remain immutable. Common claims preserve variants; each topic assessment uses its own qualifying facts
P0	`src/lib/analysis/call.ts` retries; gateway defaults allow very large outputs	A dense domain can truncate and repeatedly fail at the same boundary	Bound both input and output, split deterministic source/record manifests, reconcile only boundary candidates
P1	Retry layers exist in SDK, analysis wrapper and worker; cost estimates precede work	Attempts, search, repair and uncertain timeout spend can exceed visible estimates	One persisted update allowance with attempt ledger and reservations; consolidate retries
P1	Retrieval chunks and readable page counts are not equivalent to complete reading	Dashboard overstates evidence coverage, especially tables and scans	Show source quality and bounded-search limitations; route only affected pages through optional OCR/table handling

Priority	Current evidence	Risk	Smallest useful revision
P1	Current schema lacks explicit baseline, scenario, target scope and responsible actor	Different targets are merged, or ordinary capacity is mistaken for achievement	Small nullable fact/delivery objects, with source references; group by identity and negative constraints
P1	Current UI relies on stage maxima and latest-versus-prior document comparisons	NAP/NDC purpose differences become misleading progress arrows	Fixed source profiles; deterministic topic coverage; arrows only for comparable documentation snapshots
P1	Taxonomy editor behavior can couple codes to current placement	Moving a topic breaks identity/history	Keep permanent IDs; home and secondary placements are separate editable fields
P1	Prompt/configuration and human explanations are scattered	Humans and future coding assistants may choose an obsolete file	One entry point, one bundle, two normal templates; archives explicitly excluded from inputs

What the model should and should not receive

Extraction receives a short role/task prompt, compact global topic vocabulary, detailed local routing definitions, enabled label choices, a strict output schema, source identity and a bounded set of page/context units. The global vocabulary lets a water passage found during agriculture retrieval route correctly. Detailed definitions of every optional facet need not accompany every request.

Organization receives source-linked records and candidate context from the selected corpus. It needs quotes, scope, time, conditions and evidence references to avoid false merging. It does not need the full PDFs again, earlier scores, the team handbook, old prompt files or arbitrary repository search results. Scoring is deterministic code after organization, not a third reasoning stage.

Neither JSON Schema nor exact quote occurrence proves that the interpretation is correct. They catch shape/provenance errors. Semantic attribution and identity errors remain a reason for conservative prompts, retained variants and representative-source evaluation. No design can promise zero hallucinations without measuring actual model behavior.

Context and cost: measured versus proposed

The proposed 8k target / 18k hard source-token limit, 12k / 24k organization-input limits, 6k output reserve and parallelism of four are initial configurable hypotheses. They are not benchmark results. Count the whole request: instructions, vocabulary, schema, data, output/reasoning reserve and margin. A character estimate is an offline check, not an exact tokenizer or provider guarantee.

The existing rough estimator uses around 19k input and 4k output per extraction. For three documents and 16 domains, the ordinary pattern is 48 extraction units plus 16 organization units, before retries/splitting. A rough 1.08 million input / 256k output estimate is illustrative, not observed usage or a price quote. Savings depend on retrieval, actual outputs, repeated prefixes, model choice, batching and reuse. Measure actual billed usage before asserting a saving.

Keep domain retrieval for the first update. Section-first extraction is a benchmark option, not an automatic improvement: it can reduce repeated passages but needs reliable section segmentation, cross-section context and another deduplication burden. A single giant country prompt loses source-local repeatability and makes incremental updates expensive. Two logical stages with bounded parallel units remain the best fit here.

Edge cases and the intended behavior

Case	Required behavior
NDC says 100,000 ha; NAP says 80,000 ha for the same scope/year	Keep both values as unresolved variants; show a conflict badge, not an arbitrary score penalty
One value is additional area, another total area	Keep different metrics; do not declare a contradiction
Conditional and unconditional targets	Preserve both conditions; do not average or treat them as disagreement
Baseline 2010 versus 2015, gross versus net, different scenarios	Negative merge constraints; explicit compatible relationship only
Existing 60,000 ha network supports an irrigation plan	Context supports relevance; it is not target achievement
Finance allocated but no physical work reported	Funding state stays visible; it does not establish physical implementation
Planned law with named actor and date	Specific verifiable deliverable may qualify without a quantity; no fabricated numerical threshold
National ministry named elsewhere in the document	Do not lend it to every intervention without an explicit source connection
One programme contains several different projects	Shared programme name alone does not prove identical intervention
A~B and B~C look similar but A and C differ in scope	No transitive merge without checking the whole identity set
Claim mentions agriculture and health	Each topic needs its own evidence-supported predicates; no automatic inheritance of the claim-wide band
Cross-domain record arrives after one domain job	Frozen barrier includes all contributing extraction units; late completion creates a newer snapshot
New document arrives while an update runs	Put it in the next manifest or coalesce before freeze; never mutate the in-flight corpus
Output truncates in a dense table	Split source-unit manifest, keep headers/footnotes as context, deduplicate and reconcile boundaries
Table row has a number but its unit is in the header	Include separate cited header context; no invented stitched quotation
Source is unreadable, scanned or missing a page	Optional targeted OCR; otherwise incomplete coverage, never a confident zero
Retrieval finds no evidence after all planned searches	No evidence found in this bounded supplied corpus; not proof of national policy absence
Document includes instructions to the assistant	Source text and metadata are data only; ignore embedded instructions

Case	Required behavior
Relevant statement has no suitable topic	Keep unmapped evidence, original statement type and quotes; no invented Other domain
Generic education or tourism policy unrelated to climate/disaster	Omit as outside scope; sector names alone do not make it eligible
Water is a stated priority for one country	Priority lens changes focus; common-topic denominators and weights stay fixed
Landlocked country and fisheries topic	Do not infer non-applicability: inland fisheries exist; silence never creates N/A
Newer NAP edition and older NDC	Source profile selects eligible editions by role; selection does not assert one instrument repeals the other
Older expired target absent from new edition	Retain history; do not silently carry it into the current corpus or infer withdrawal
Missing NAP in an NDC+NAP comparison	Mark source profile incomplete; retain positives and express unresolved potential coverage
Model/prompt changes improve extraction	New baseline or refresh marker; no upward policy-progress arrow
One topic has several commitments but only one detailed action	Topic may reach band 3; disclose 1 of N detailed responses in its drawer
User hides a claim after publication	Preserve raw data and canonical snapshot; mark customized view. Comparison must not silently mix personal overlays
Budget exhausted or provider timeout	Retain positives, publish partial status; reserve uncertain spend and retry within the same allowance
Organization split cannot resolve boundary identity	Keep separate provisional groups and incomplete grouping status; no silent evidence loss

Release evaluation, without a human moderation queue

Use fixed fixtures plus a representative source set containing scanned pages, tables, multilingual text, repeated targets, conflicting targets and different document editions. Automated checks cover schema, references, quote integrity, accounting, topic ownership, deterministic scores, idempotence and snapshot ordering. Evaluate semantic recall and false merges against a fixed benchmark; do not substitute a second model's confidence for ground truth.

Choosing an evaluation set and maintaining software tests is product development. It does not introduce a manual approval gate for each upload or publication. The production path either publishes completed evidence or clearly labelled partial evidence automatically. Current pack tests validate reference contracts and scoring logic; they do not establish live extraction accuracy.

Future extensibility boundary

Keep interfaces for source preparation, retrieval, extraction, organization and scoring. Store facts and provenance, not only display prose. A future OCR service, document type, model provider or analyst tool can implement an interface without replacing the taxonomy or creating extra routine reasoning stages. Add modules when an observed user need justifies them, not because all possible tools could be supported.

Primary references behind the recommendations

Checked 19 October 2026. These sources inform the design; the proposed coverage metrics and four bands are this project's method, not an official UN score.

• OpenAI: Structured Outputs. Strict schemas improve format adherence but do not eliminate content mistakes. Explicit null/empty paths and generated enum contracts help avoid forced guesses.

• OpenAI: Prompting. Clear task instructions, examples and versioned evaluation support repeatable prompt changes. Our file allowlist is an application design decision.

• OpenAI: File search. Retrieval result limits trade token/latency cost against information coverage. We therefore distinguish code-logged searches from evidence completeness.

• OpenAI: Prompt caching. Repeated prefixes can benefit from caching. Static instructions/vocabulary precede variable data; savings must be measured for the selected model and actual traffic.

• UNFCCC: Technical guidelines for the national adaptation plan process. NAP planning is country driven and non-prescriptive. This supports showing priorities alongside a common documentation comparison rather than claiming all countries need identical sector portfolios.

• UN Statistics: 2025 SDG progress assessment technical note. SDG progress assessment uses indicator series, targets and explicit data-availability rules. Document extraction alone does not provide that outcome evidence; the proposed arrows describe documentation change only.

• UNDRR: Sendai Framework. The disaster-risk frame extends beyond climate hazards. Keep all-hazard disaster records distinct from climate attribution and avoid treating NDC/NAP completeness as a full national DRR assessment.

No vendor pricing, universal token limit or guaranteed accuracy/cost reduction is assumed. Re-check model capabilities and limits during implementation. The current runtime source archive is the authority for application behavior; external documentation does not override what that code actually does.